

PROJECT APPRAISAL

Module 9

Risk Analysis in Project Appraisal



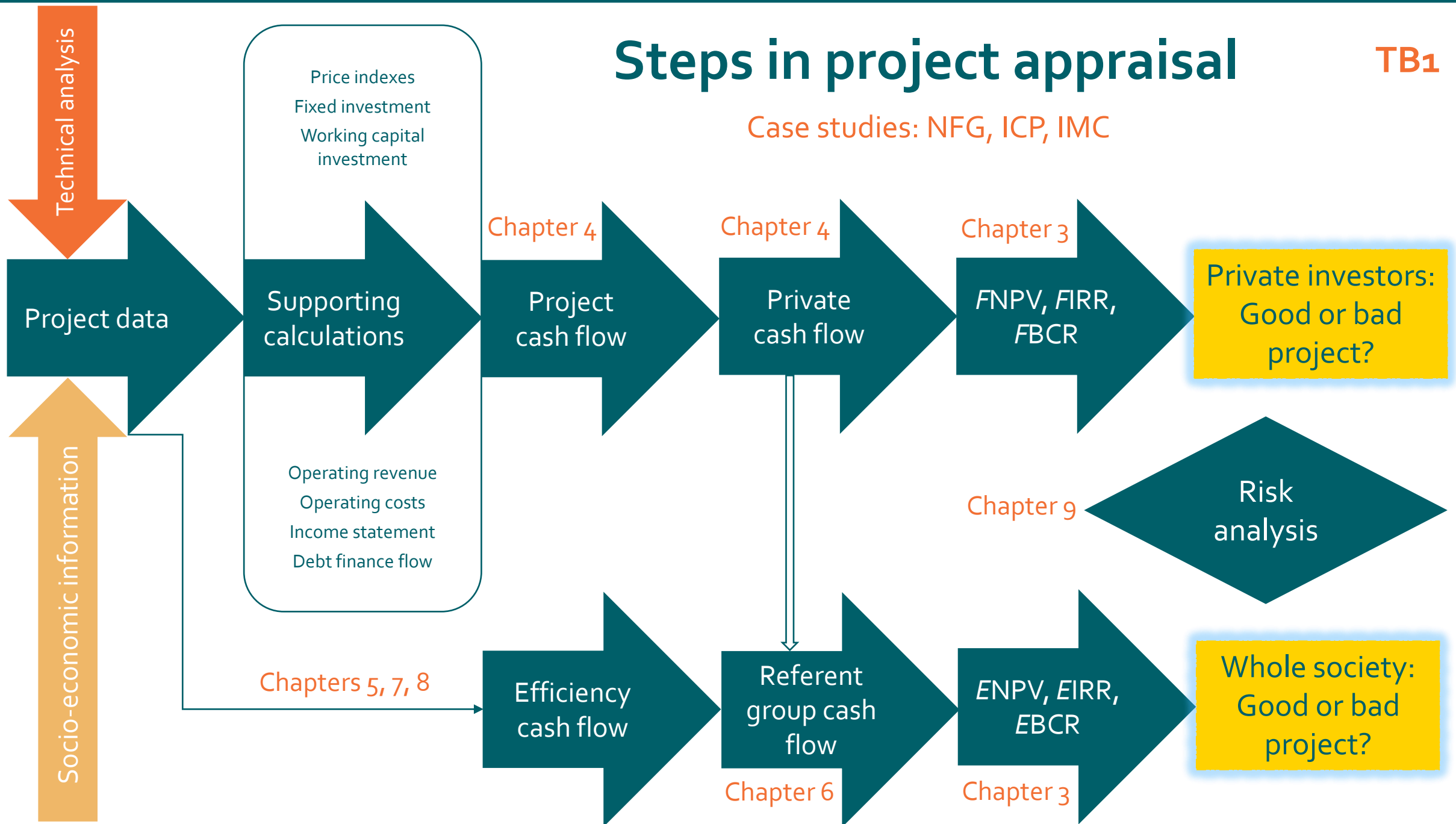
Outline – Module 1

- Introduction: Risk and uncertainty
- The value of information
- Sensitivity analysis
- Scenario analysis
- Risk modelling: Use of probability distributions
- Risk modelling: Monte Carlo simulation
- Qualitative risk analysis: Risk matrix
- An illustrative example: PEO project

Steps in project appraisal

TB1

Case studies: NFG, ICP, IMC



Introduction

Risk and uncertainty

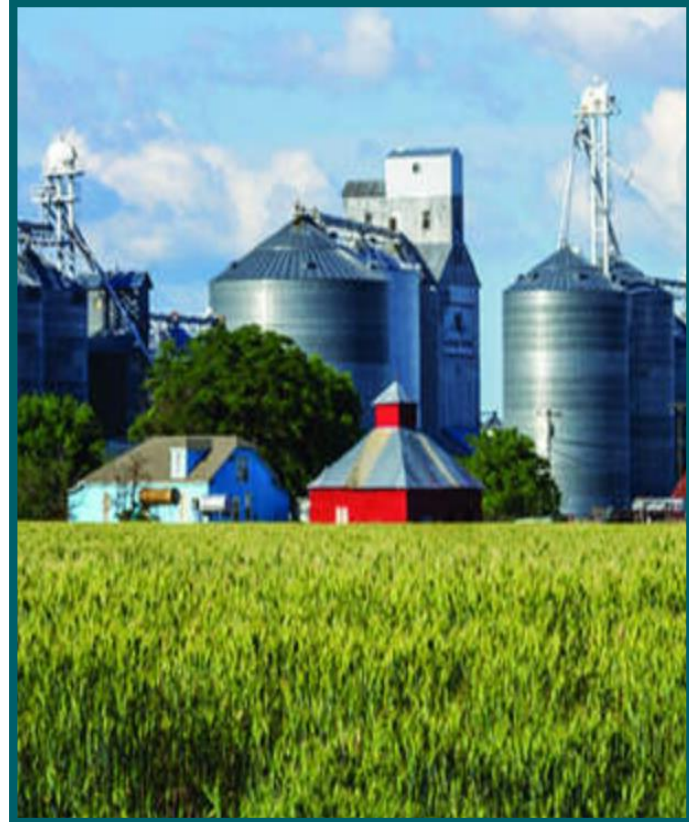


Risk and uncertainty

- In project appraisal, the value of none of the variables can be observed, but rather has to be predicted.
- Up to now, we have implicitly assumed that all costs and benefits are known with *certainty*.
- However, the future is *uncertain*, we do not know what the future values of a project's costs and benefits will be.
- Uncertainty implies that there is more than one possible value for any project's costs and benefits.
- Under the range of possible values, our decision about the project will involve taking a *risk*.

The value of information

Sensitivity analysis



The value of information

- If the project is found to have a positive NPV and would have gone ahead in the absence of the analysis, the CBA has contributed nothing.
- If the project would not have gone ahead and is found to have a negative NPV, again nothing has been gained.
- Information has value when it contributes to overturning a potentially wrong decision:
 - Avoid rejecting a good project (Type I Error)
 - Avoid undertaking a bad project (Type II Error)

The value of information

We must compare the expected value of information with the cost of collecting it in deciding what information to collect.

Prior decision	Accept the project	Reject the project
CBA Result = Good (NPV > 0)	Value of information = 0	Value of information > 0
CBA Result = Bad (NPV < 0)	Value of information > 0	Value of information = 0

Sensitivity analysis

- Additional information should be obtained only when it has a corresponding benefit in terms of the likely quality of the decision to be made.
- If the estimate of project NPV is likely to remain positive (or negative) over the feasible range of values of a variable, there may be no benefit to be gained from a more precise estimate.
- Where the value a variable takes within its feasible range can affect the outcome, it may be worth acquiring additional information.

Sensitivity analysis

- In the sensitivity analysis, we examine how changes in the values of one variable, holding constant the values of all other variables, affect the project outcome.
- Sensitivity analysis is also known as “what-if” analysis.
- In Excel, it is easy to create simple **one-way** and **two-way** tables that show how changes in the value of a certain variable impact on the NPV of the project.
- The generation of the one-way or two-way tables requires that the cells for the outcomes are properly linked to the variables in the table of parameters.

Sensitivity analysis

- We can also conduct **break-even** analysis for each of the key variable (i.e. switching value, threshold value): the value of a variable at which $NPV = 0$, or IRR is equal to some target value.
- The break-even values for each variables will provide a qualitative assessment of the extent to which the project is responsive to changes in the values of the key variables.
- If the threshold value is within the range of feasible estimates, there may be a case of gathering additional information, but otherwise perhaps not.

Sensitivity analysis

Road usage benefits	Construction costs (\$)		
	75%	100%	125%
High	50	40	30
Medium	47	36	25
Low	43	32	20

Note: \$ millions at 10% discount rate.

Sensitivity analysis

Road usage benefits	Construction costs (\$)		
	75%	100%	125%
High	50	40	-30
Medium	47	36	-25
Low	43	32	-20

Note: \$ millions at 10% discount rate.

Sensitivity analysis

- **Importance of the key variables**

- Based on the results of the sensitivity analysis with all of the key parameters, we can identify and rank the set of key variables that will have the most impact on the NPV of the project.
- The relative importance of a variable depends on how much the NPV changes due to a unit change in the key variable.
- We can calculate the percentage change in the NPV due to a one percentage change in the value of the key variable.
- The selection of the important key variables will also depend on the experience, knowledge and judgment of the analyst, based on analysis from other comparable projects.

Sensitivity analysis

- **Limitations**

- The number of variables that sensitivity analysis can handle.
- The different variables are assumed to be independent and unable to model the important correlations among variables.
- The absence of probability distributions for the key variables.
- It does not contain any information about the likelihood of the project outcomes.

Scenario analysis

- Based on the set of key variables that have been identified through the sensitivity analysis, we often describe three scenarios: best case, expected case, and worse case.
- Each scenario is based on a combination of possible values from the set of key variables.
- The analysis is not restricted to three scenarios, and could specify five or more scenarios.
- Four or five variables should be sufficient for practical purposes.

Scenario analysis

- If the NPV of the project is negative in the best case scenario, it is reasonable to reject the project.
- If the NPV of the project is positive in the worst case scenario, we should accept the project.
- In practice, the results may not be so clear cut.
- Scenario analysis also suffers from the other shortcomings identified in sensitivity analysis such as the likelihood of the different scenarios, the correlations among the variables, ...

Risk modeling

Use of probability distributions



Discrete probability distributions

- Once a probability distribution for a project input variable has been chosen, the analyst is able to calculate the expected value of the variable, and to use this in the cash flow for the project rather than the point estimate.

	Road Construction Cost (C) (\$ millions)	Probability (P) (%)	$E(C) = P \times C$ (\$ millions) = 95	NPV (\$ millions)	$E(NPV)$ (\$ millions) = 41
Low	50	20	10	86	17.2
Best Guess	100	60	60	36	21.6
High	125	20	25	11	2.2

Note: Present Value of net benefits, excluding road construction cost, is assumed to be \$136 million.

Joint probability distributions

- If there is uncertainty about the level of more than one project input or output, the probability distribution for the outcome will depend on the aggregation of probability distributions for the individual variables into a joint probability distribution.
- In aggregating probability distributions, we must distinguish between correlated and uncorrelated variables. For examples:
 - Road maintenance cost and user benefit (correlated)
 - Local cost and export price (uncorrelated)

Joint probability distributions

There is 20% chance of road maintenance costs being \$50 and road user net benefits being \$125, a 60% chance of road maintenance costs being \$100 and road user net benefits being \$125, and 20% chance of road costs being \$125 and road user net benefits being 205. The expected NPV is equal **\$35** million.

Joint probability distribution: correlated variables

	Probability (P) (%)	Cost (\$ millions)	Benefits (\$ millions)	Net Benefits (\$ millions)
Low	20	50(10)	70(14)	20(4)
Best Guess	60	100(60)	125(75)	25(15)
High	20	125(25)	205(41)	80(16)
(Expected Value)		(95)	(130)	(35)

Joint probability distributions

Joint probability distribution: uncorrelated variables

Level	Probability(P) (%)	Cost (\$ millions)	Benefits (\$ millions)
Low (L)	20	50	70
Best Guess (M)	60	100	125
High (H)	20	125	205

Combination	Joint Probability	Net Benefit (\$)
LC-HB	$0.2 \times 0.2 = 0.04$	155(6.2)
LC-MB	$0.2 \times 0.6 = 0.12$	75(9.0)
LC-LB	$0.2 \times 0.2 = 0.04$	20(0.8)
MC-HB	$0.6 \times 0.2 = 0.12$	105(12.6)
MC-MB	$0.6 \times 0.6 = 0.36$	25(9.0)
MC-LB	$0.6 \times 0.2 = 0.12$	30(3.6)
HC-HB	$0.2 \times 0.2 = 0.04$	80(3.2)
HC-MB	$0.2 \times 0.6 = 0.12$	0(0.0)
HC-LB	$0.2 \times 0.2 = 0.04$	-55(-2.2)

$$\begin{aligned}
 E(NPV) &= 6.2 + 9 + \\
 &0.8 + 12.6 + 9 + \\
 &3.6 + 3.2 + 0 - 2.2 \\
 &= \text{\textbf{\$42.2 million}}
 \end{aligned}$$

Continuous probability distributions

- A continuous probability distribution assigns some probability to the event that the outcome lies in any particular segment within the range of the distribution.
- The analyst must decide what type of probability distribution best describes the uncertain variable.
- Some public sector jurisdictions offer the analyst advice about the appropriate form of distribution to be used.
 - Poisson or Binomial Distribution for a discrete variable
 - Normal Distribution for a symmetric and continuous variable
 - Triangular Distribution for an asymmetric and bounded variable
 - Uniform Distribution if no information about the value of the variable, apart from its likely bounds, is available.

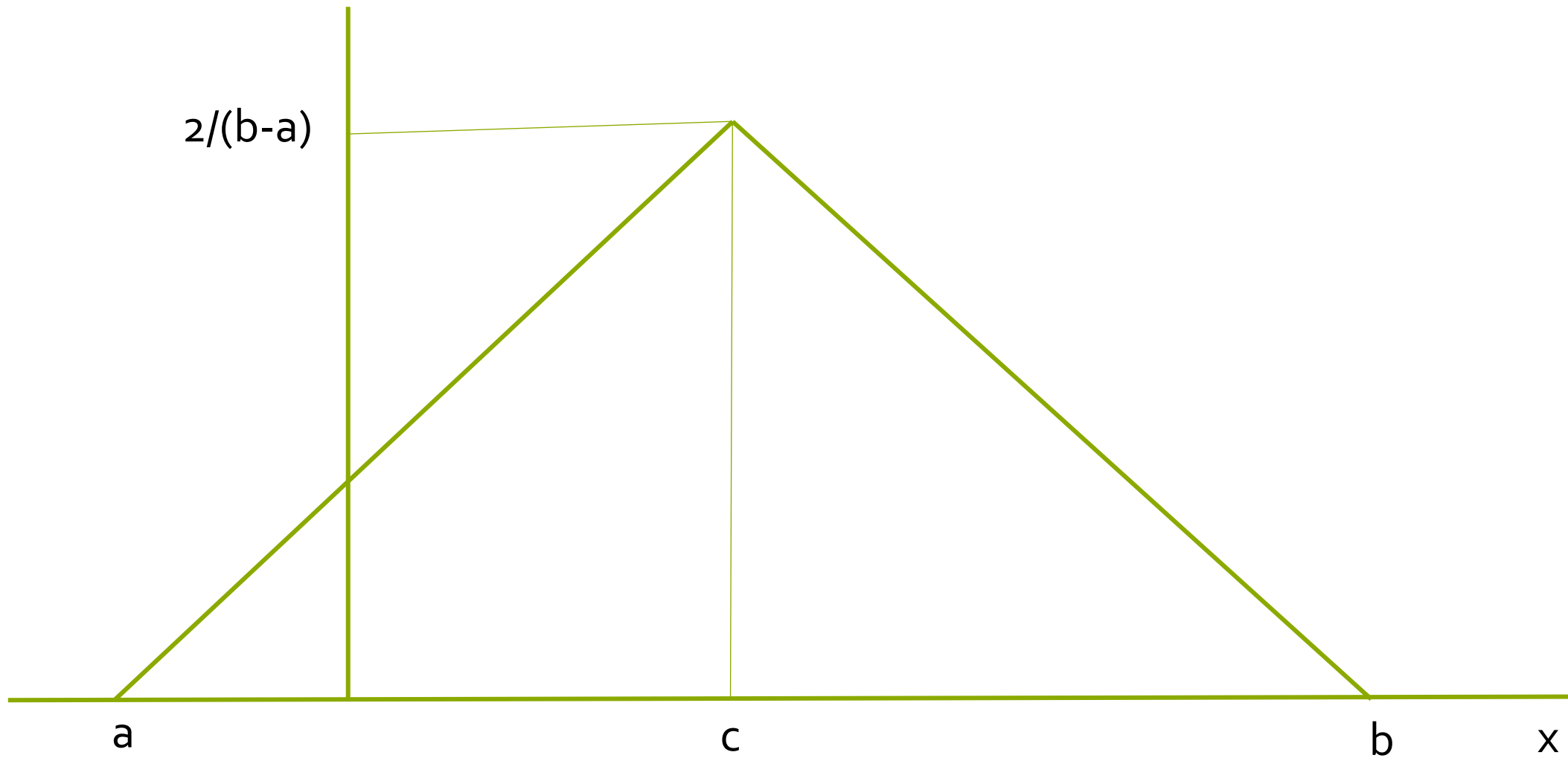
Continuous probability distributions

- Risk modelling programs, such as **Crystal Ball**, offer the analyst a whole range of distributions to choose from.
- If the analyst has a reasonable set of past observations on the uncertain variable in question, but is unsure as to what type of probability distribution best describes that data, the program offers the **bestfit** option to identify the most appropriate probability distribution.
- Quite often, there may be **no** reliable historical information about the variable in question and **no** information beyond the range of values from the sensitivity analysis.

Continuous probability distributions

- In such a case, a **Triangular Distribution** could be the most appropriate choice.
- The Triangular Distribution is described by a high (H), low (L) and best-guess (B) estimate.
- Triangular Distribution is particularly useful because information about the distribution of the variable has to be obtained from **experts** in areas relevant to the project.
 - What do you regard as the most likely value this variable will take?
 - What is the lowest value the variable could take?
 - What is the highest value this variable could take?

Continuous probability distributions



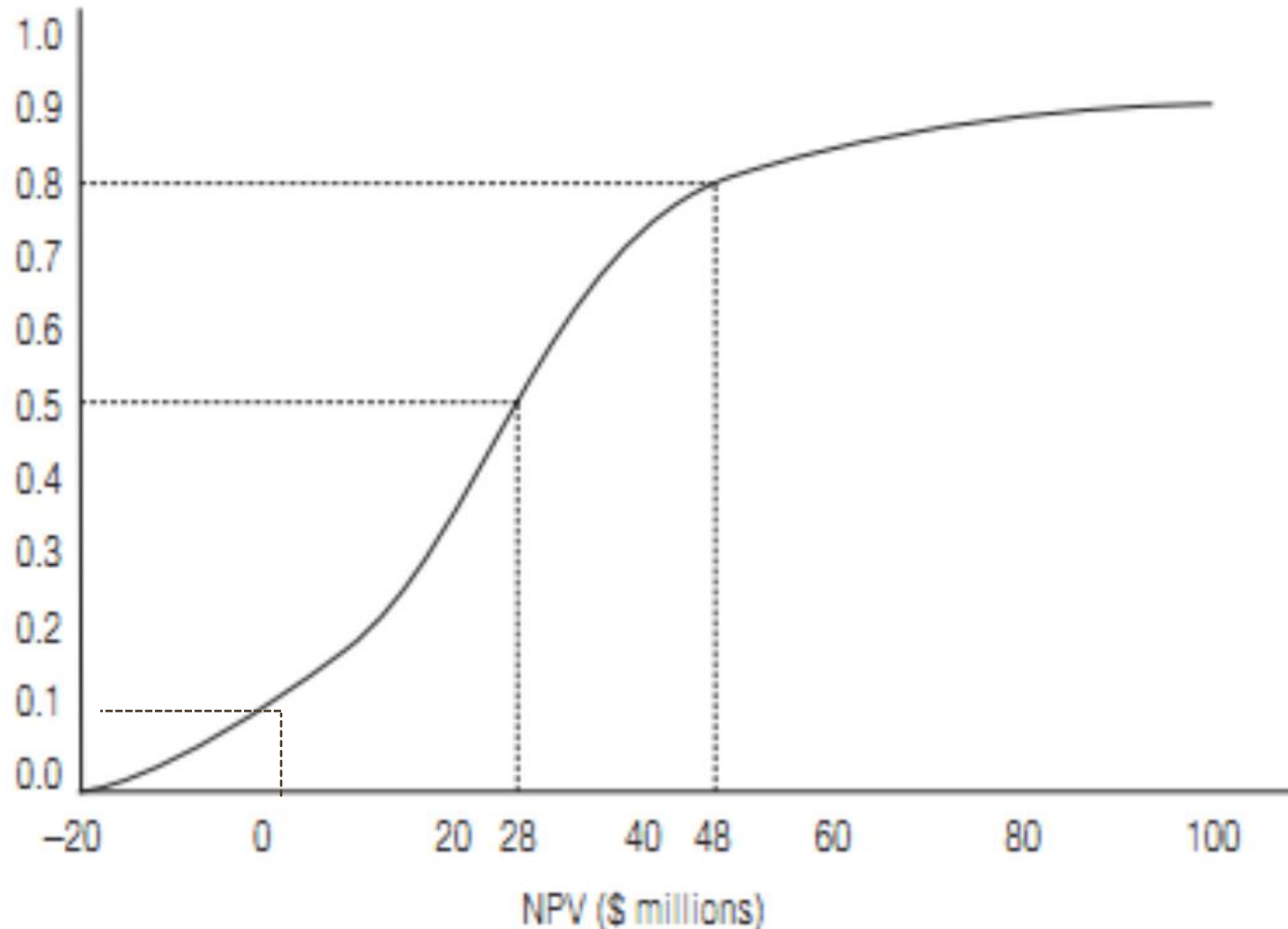
The expected value (Mean) of x is given by $(a + b + c)/2$, the Median value lies between the Mean and the Mode c).

Continuous probability distributions

- Selected variables involved in the project CBA can be described by probability distributions, and, using a program such as Crystal Ball, a probability distribution can be derived for the project's NPV.
- A useful format for presenting this information is a cumulative probability distribution.
- The advantage of presenting the results of a risk analysis in this way is that it provides the relevant information to the decision-maker in a user-friendly, and easy to interpret graphical format.

Continuous probability distributions

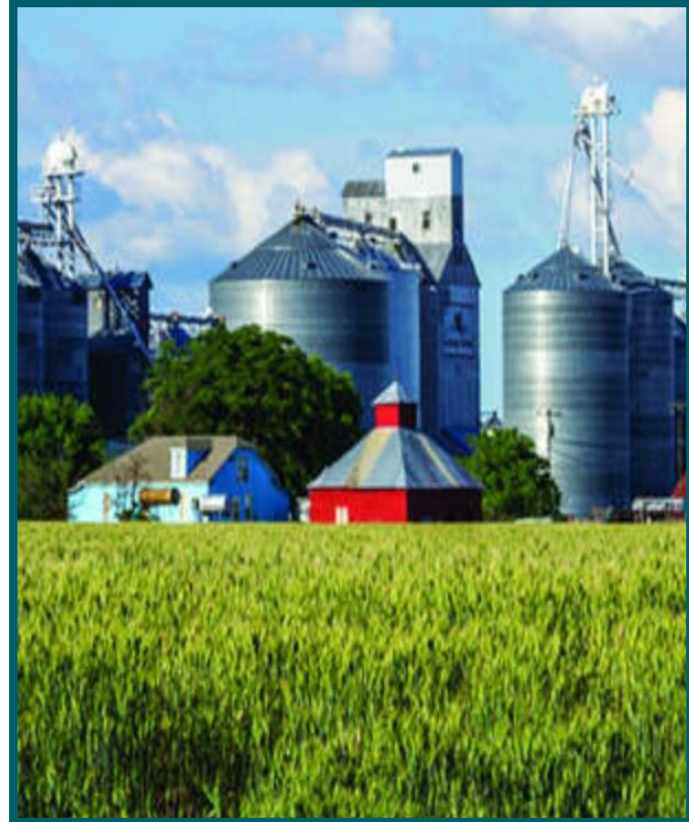
Cumulative Frequency (%)



The distribution shows the probability of NPV being less than the value shown on the horizontal axis. For example, there is a 10% chance of NPV being negative. The chance of NPV lying between zero and +\$28 is calculated by subtracting the probability of a negative value from the probability that it will be less than \$28, to give 40%. There is virtually no chance of NPV exceeding \$100.

Risk modeling

Monte Carlo Simulation



Monte Carlo simulation

- MCS is a natural extension of sensitivity analysis and overcomes the two main major shortcomings of sensitivity analysis (i.e. probability distributions of the key variables and the potential correlations among the key variables).
- With MCS, we can specify probability distributions for the key risk variables, specify correlations that exist among the risk variables, and can model a reasonably large number of variables in a sensitive manner.
- By specifying the probability distributions for the variables, we obtain the probability distributions for any desired outcomes such as NPV of the project from different points of views..

Monte Carlo simulation

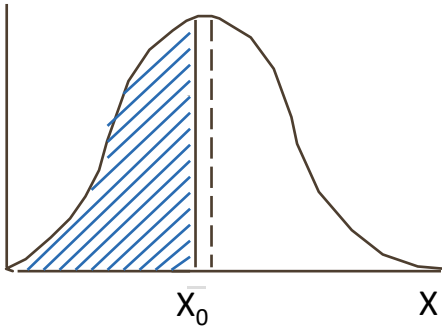
- In conducting a MCS, we undertake the following steps:
 - 1) Mathematical model: project appraisal spreadsheet
 - 2) Identify variables that are sensitive and uncertain
 - 3) Define uncertainty
 - Specify probability distributions
 - 4) Identify and define correlated variables
 - Positive or negative correlation
 - Strength of correlation
 - 5) Run the simulation
 - 6) Analysis of results (Summary statistics, probability distributions of desired outcomes, and other techniques)

Monte Carlo simulation

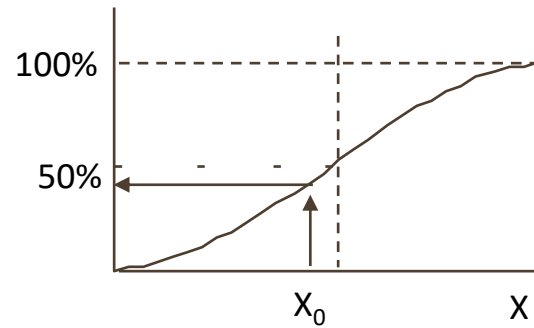
- **Step 1:** Ensure that the desired outcomes are properly linked via formulas to the key variables in the table of parameters.
- **Step 2:** Conduct sensitivity analysis. From a practical point of view, six to eight risk variables would be a reasonable number.
- **Step 3:** Commonly used distributions are: the uniform, triangular, and normal. Historical information, experts and secondary data from the published literature and other comparable and similar projects.
- **Step 4:** Specify the direction and strength of the correlations.
- **Step 5:** Specify the desired outcomes, number of trials, and run the simulation model.

Monte Carlo simulation

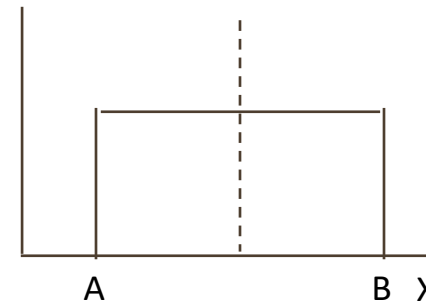
Probability



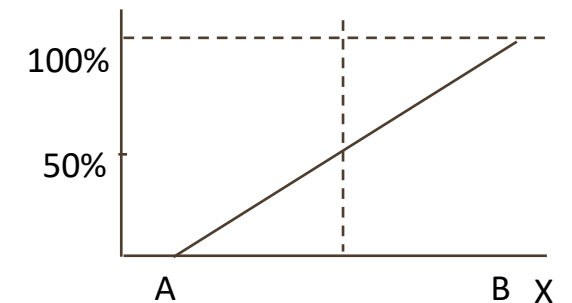
Cumulative probability



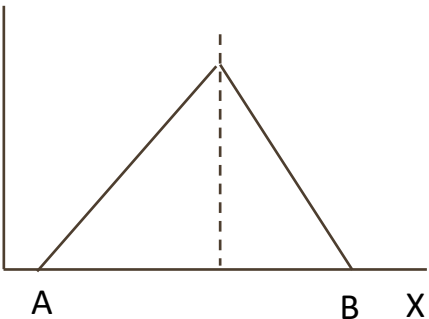
Probability



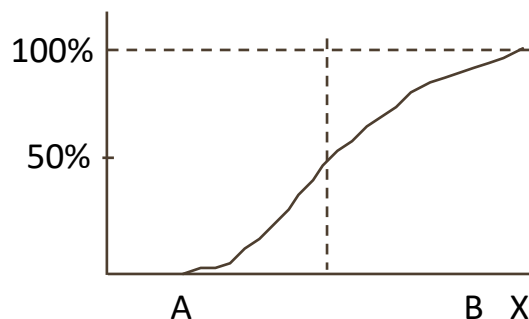
Cumulative probability



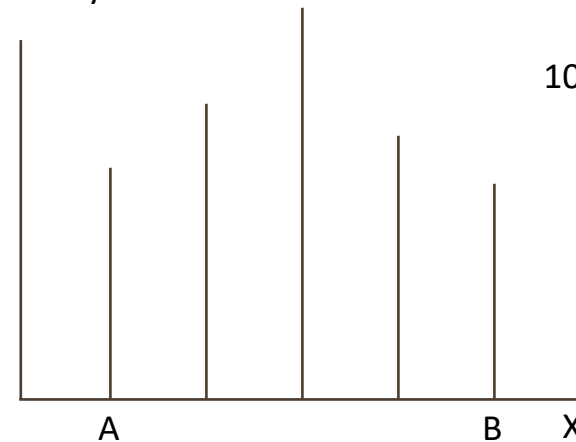
Probability



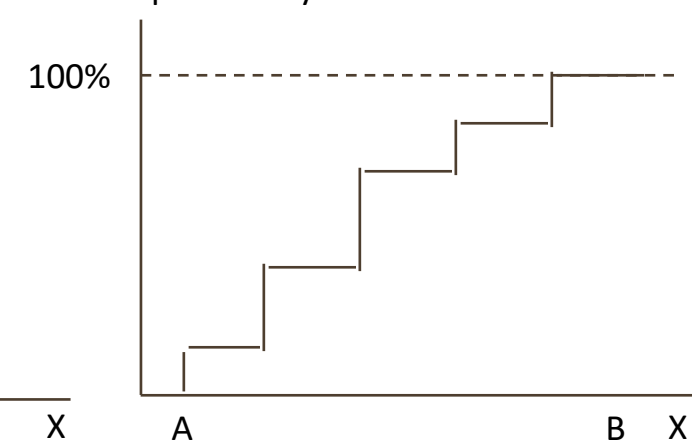
Cumulative probability



Probability



Cumulative probability

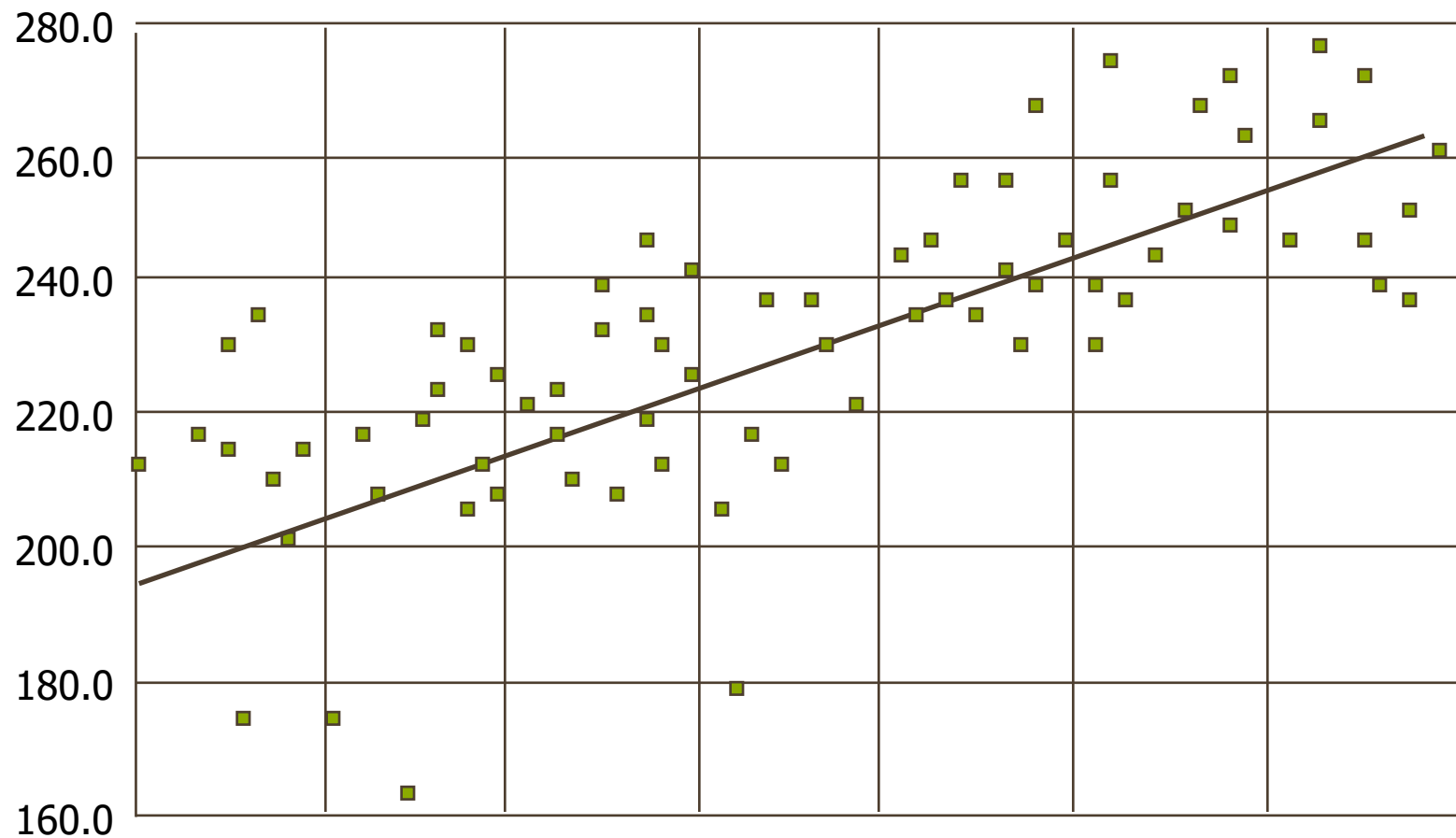


Monte Carlo simulation

Correlation between X and Y:

- Direction: Positive
- Strength: 0.5

Output Y

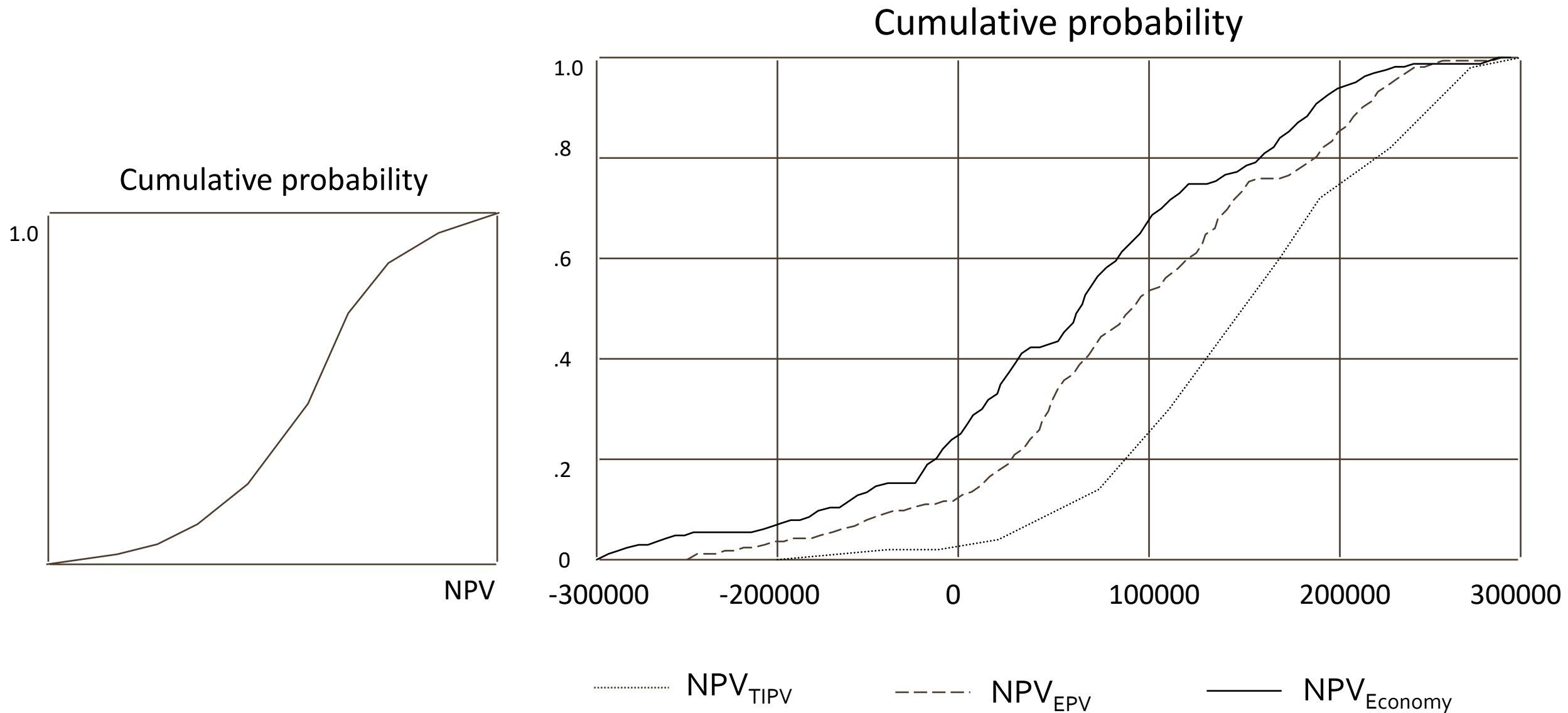


Input X

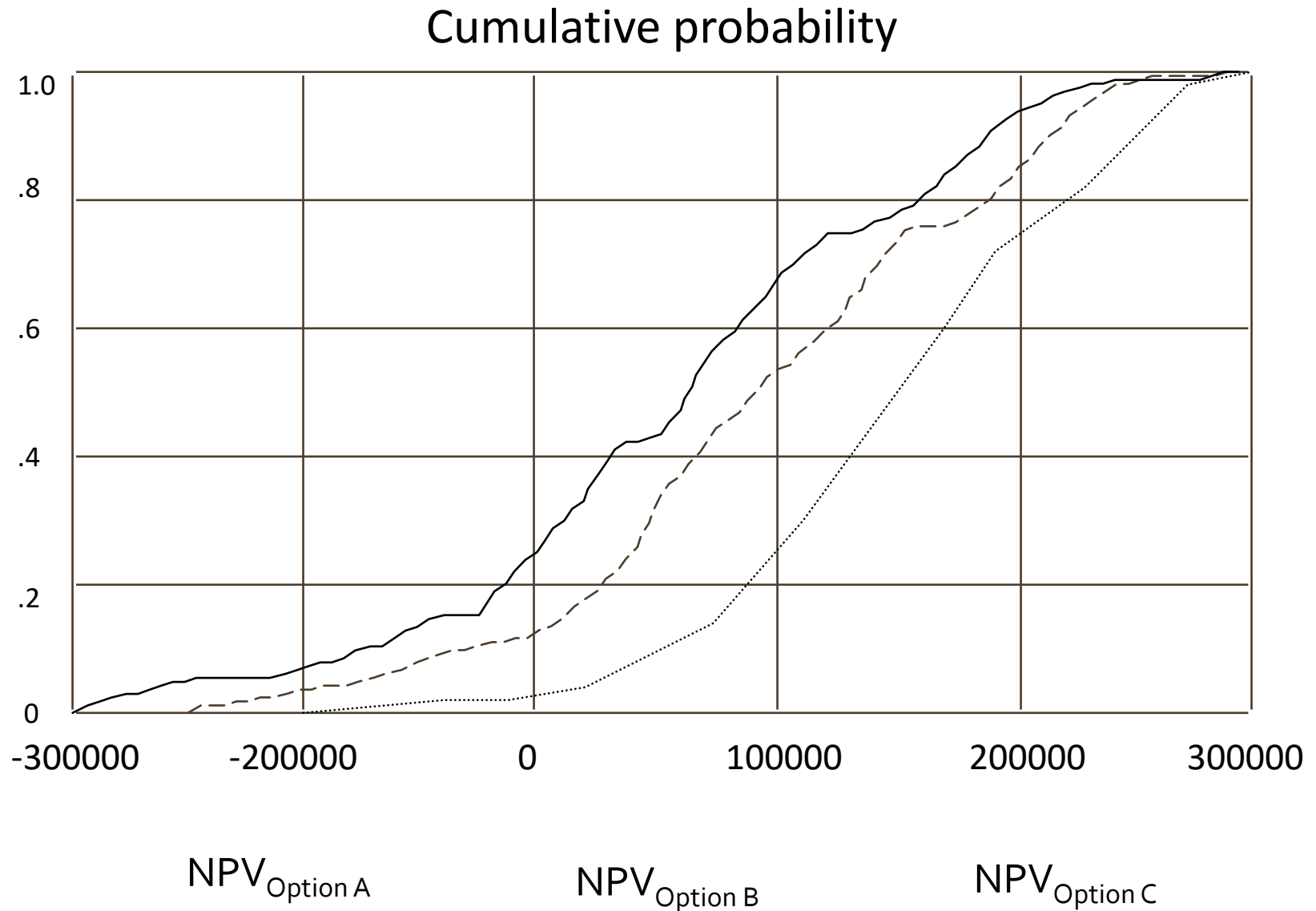
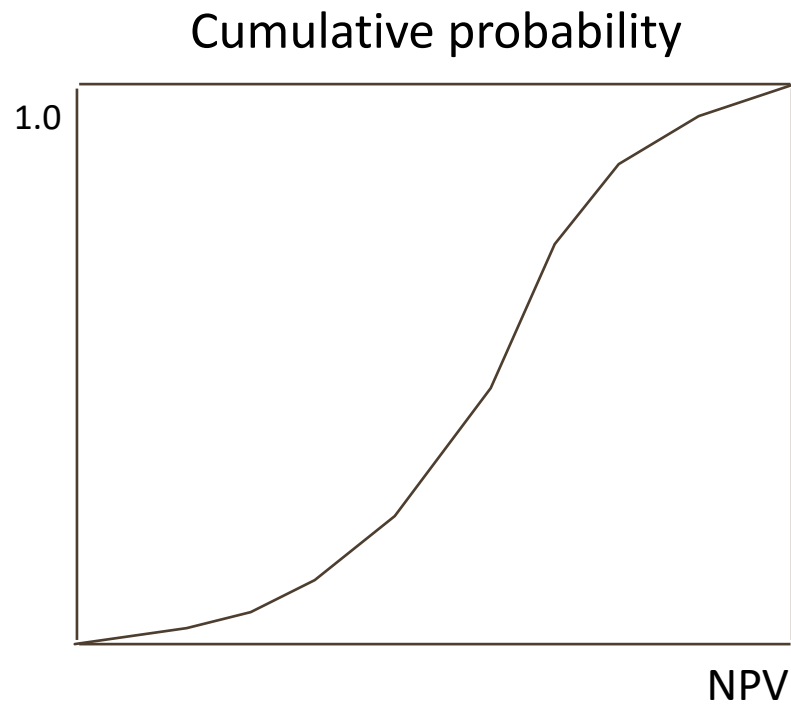
Scatter of X and Y

Regression line

Monte Carlo simulation



Monte Carlo simulation

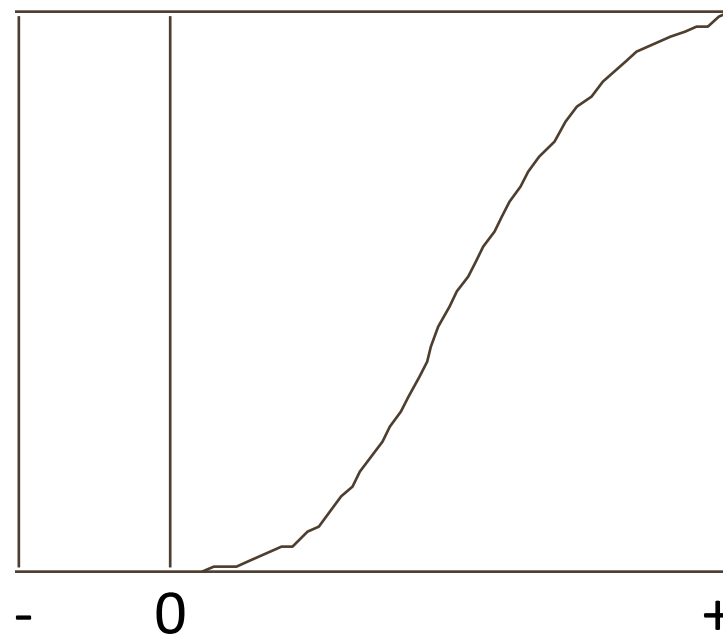


Monte Carlo simulation

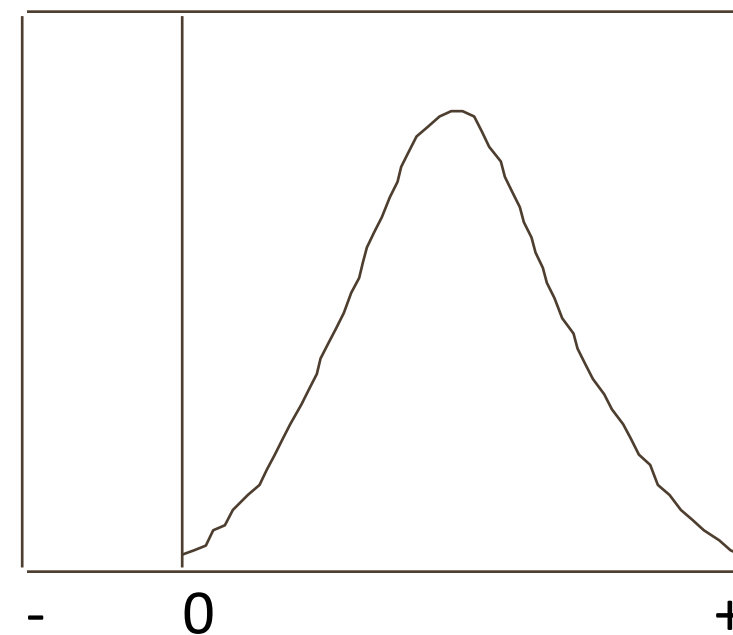
One project or
alternative

$$\text{Prob}[\text{NPV}_{\text{Project}} < 0] = 0$$

Cumulative probability



Probability



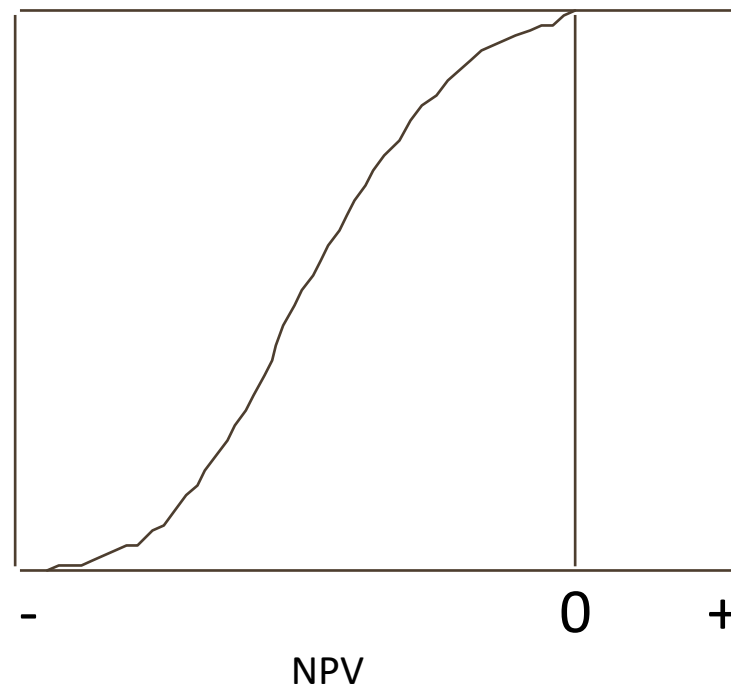
Decision rule 1: Accept the project

Monte Carlo simulation

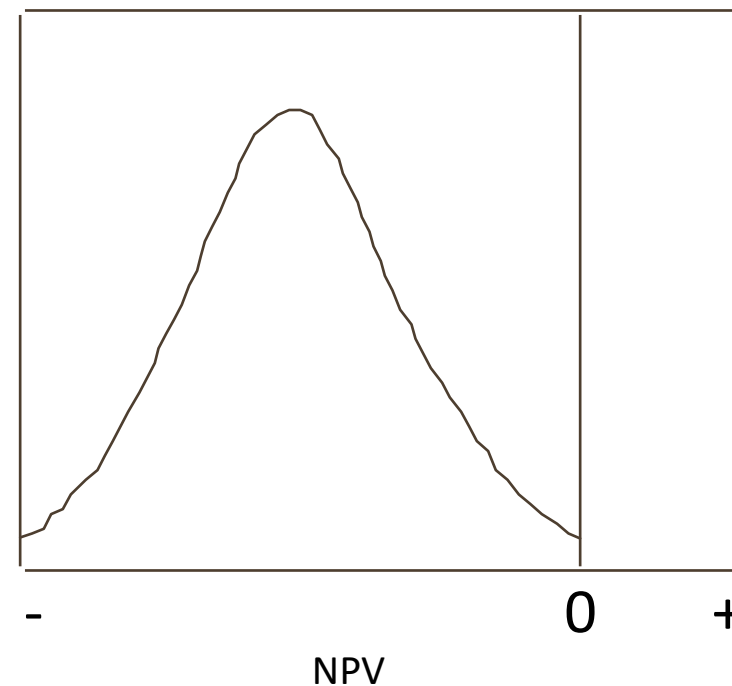
One project or
alternative

$$\text{Prob}[\text{NPV}_{\text{Project}} > 0] = 0$$

Cumulative probability



Probability



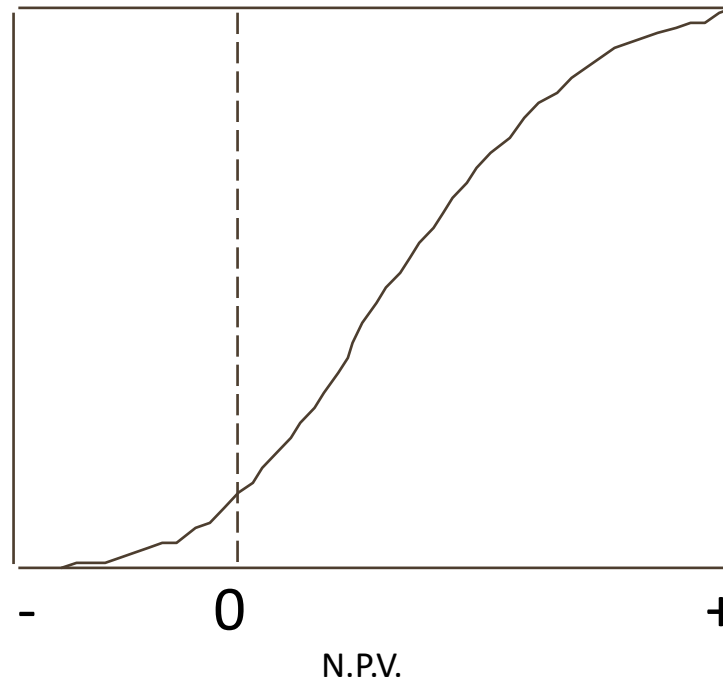
Decision rule 2: Reject the project

Monte Carlo simulation

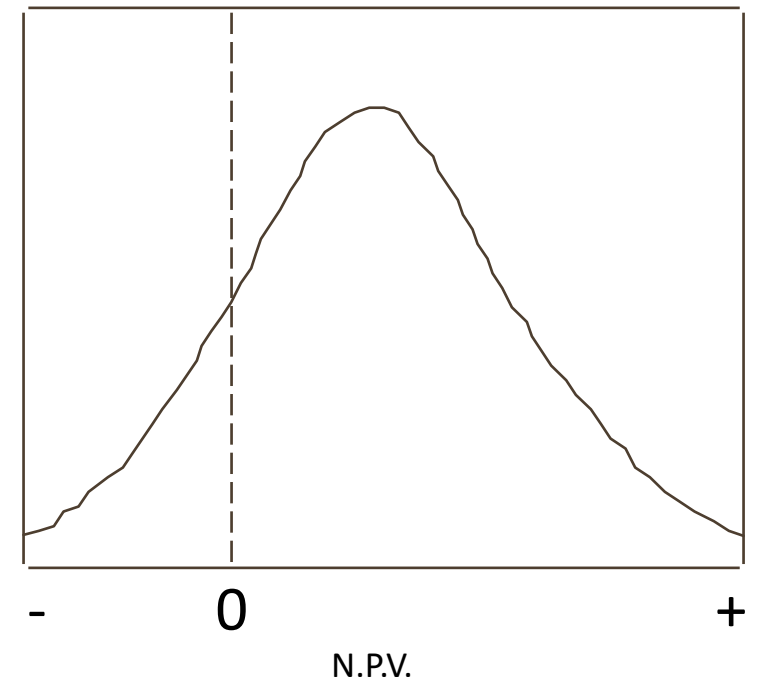
One project or
alternative

$$0 < \text{Prob}[\text{NPV}_{\text{Project}} < 0] < 1$$

Cumulative probability



Probability



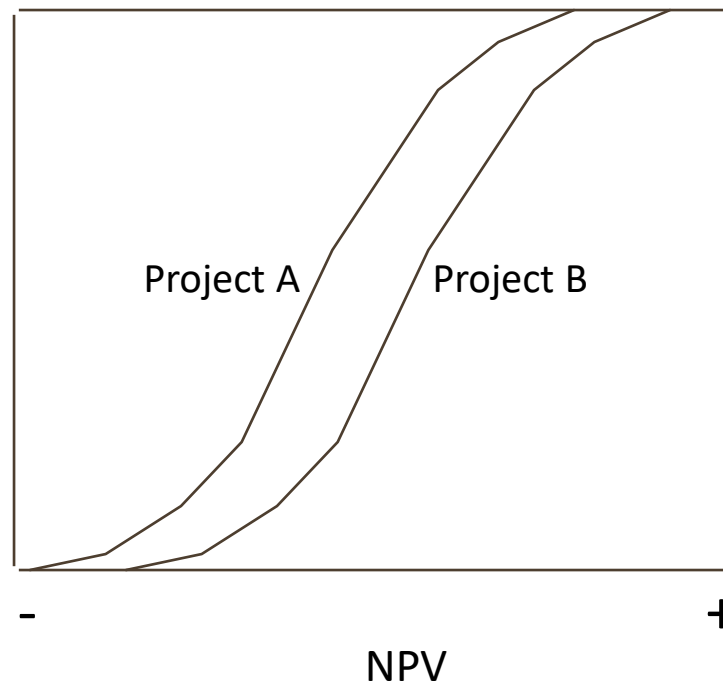
Decision rule 3: Neutral

Monte Carlo simulation

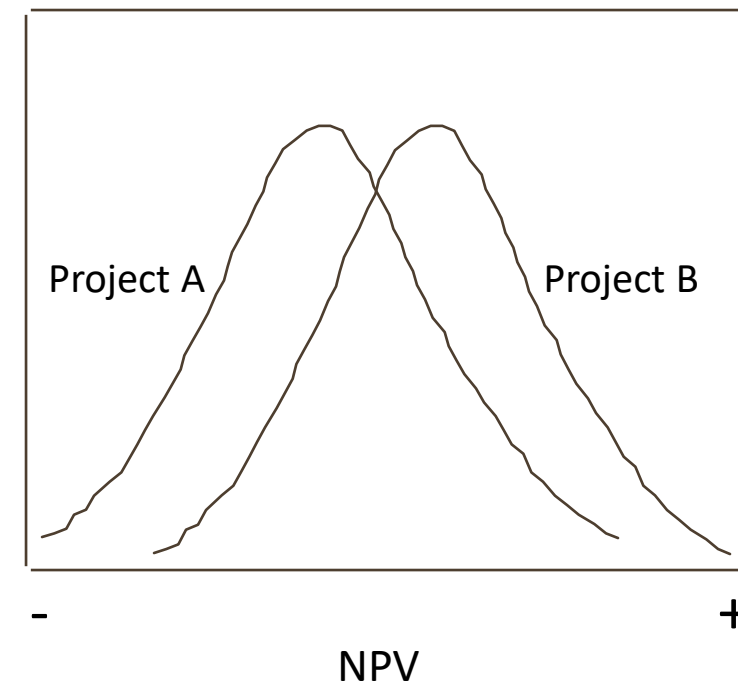
Two mutually exclusive projects or alternatives

$$\text{Prob}[\text{NPV}_A] < \text{Prob}[\text{NPV}_B]$$

Cumulative probability



Probability



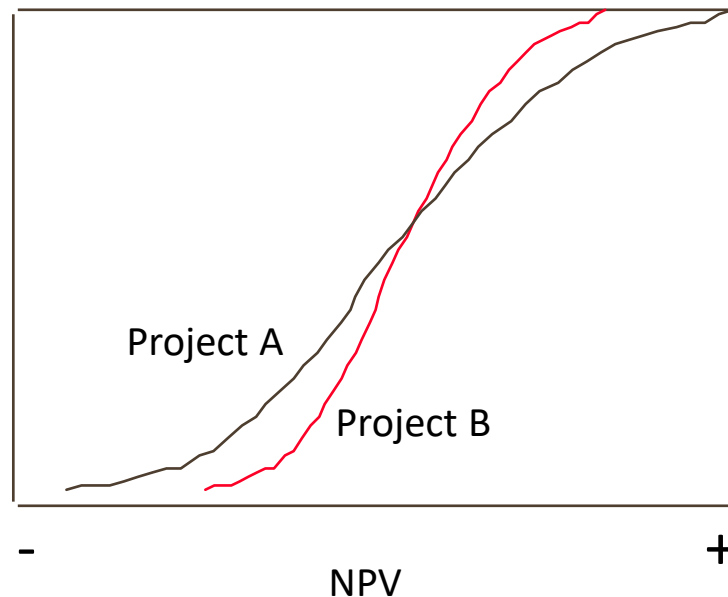
Decision rule 4: B is preferred

Monte Carlo simulation

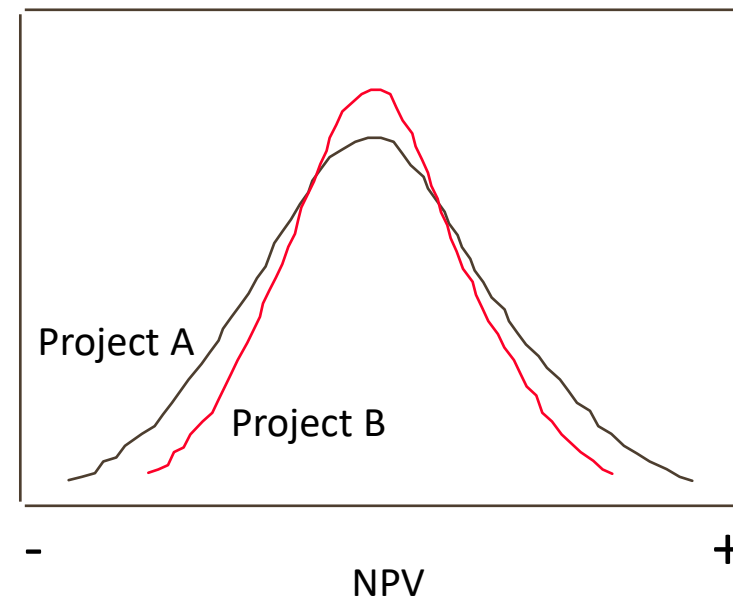
Two mutually exclusive projects or alternatives

- Risk averse: B is preferred
- Risk lover: A is preferred

Cumulative probability

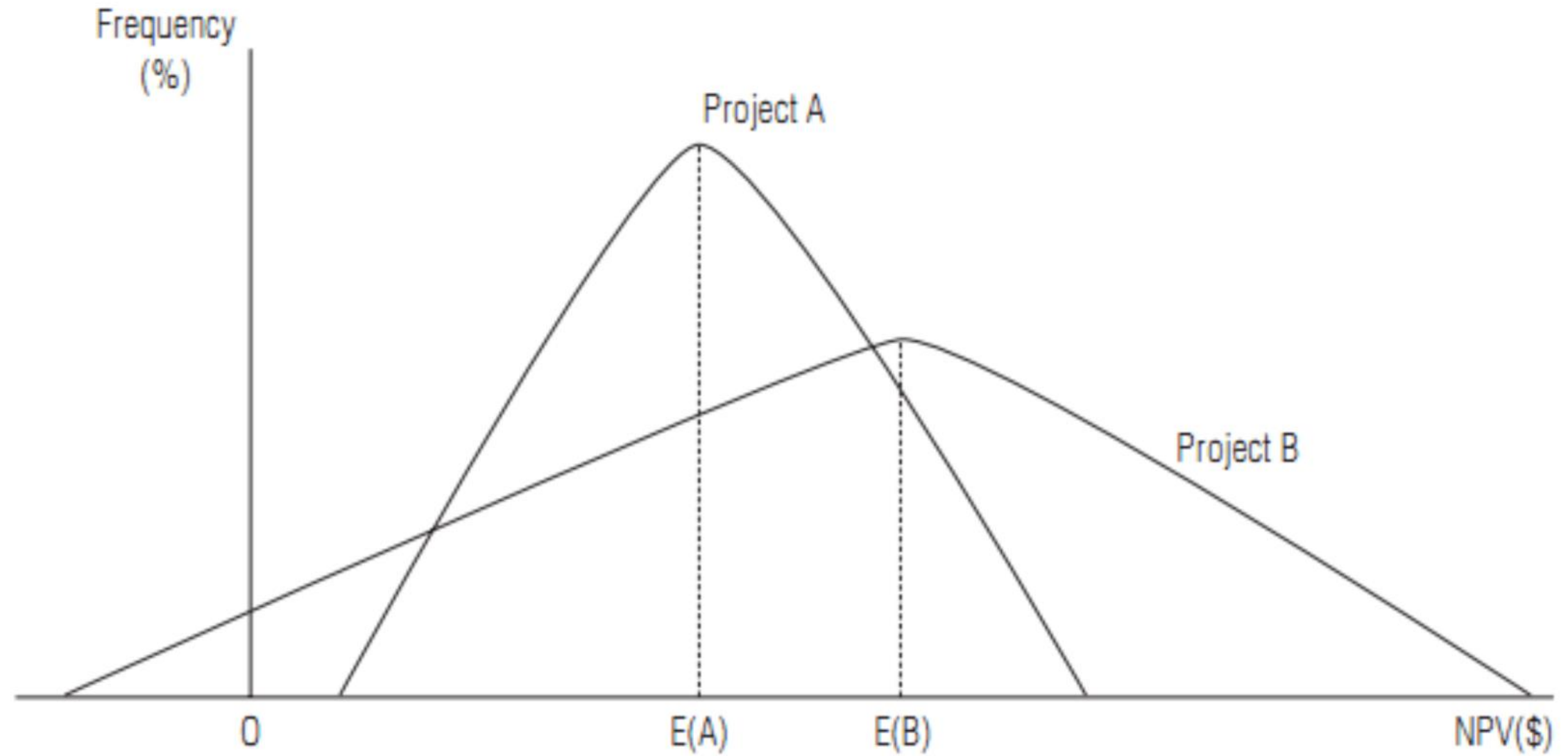


Probability



Decision rule 5: Neutral

Monte Carlo simulation



Projects with Different Degrees of Risk

Qualitative risk analysis

Risk matrix



Risk matrix

Adverse event	Variable	Causes	Effect	Timing	Effect on cash flows	Probability (P)	Severity (S)	Risk Level	Prevention and/or Mitigation measures	Residual risk
Construction delays	Investment cost	Low contractor capacity	Delay in service starting	Medium	Delay in establishing a positive cash flow including benefits materialisation	C	III	Moderate	Set up of a Project Implementation Unit to be assisted by technical assistance for project management during implementation.	Low
Project cost overrun	Investment cost	Inadequate design cost estimates	Investment costs higher than expected	Short	Higher (social) costs in the first phase of the project	D	V	Very high	The design of the project must be revised.	Moderate
Landslides	Not applicable	Inadequate site investigation	Interruption of the service	Long	Extra costs to rehabilitate the service	A	III	Low	Close monitoring	Low
Delayed obtainment of permits	Not applicable	Low political commitment; Mismanagement of the licensing procedures process	Delay in commencement of works	Short	Delay in establishing a positive cash flow including benefits materialisation	A	II	Low	Close monitoring	Low
Public opposition	Not applicable	Inadequate market strategy Underestimation of threats	Demand lower than expected	Medium	Lower revenues and social benefits	C	V	High	Early definition of an appropriate social plan; Awareness-raising activities and campaigns to raise the level of social acceptance	Moderate

Risk matrix

A Probability (P) or likelihood of occurrence is attributed to each adverse event. Below, a recommended classification is given⁷³, although in principle other classifications are possible:

- A. Very unlikely (0–10 % probability)
- B. Unlikely (10–33 % probability)
- C. About as likely as not (33–66 % probability)
- D. Likely (66–90 % probability)
- E. Very likely (90–100 % probability)

Rating	Meaning
I	No relevant effect on social welfare, even without remedial actions.
II	Minor loss of the social welfare generated by the project, minimally affecting the project long run effects- However, remedial or corrective actions are needed.
III	Moderate: social welfare loss generated by the project, mostly financial damage, even in the medium-long run. Remedial actions may correct the problem.
IV	Critical: High social welfare loss generated by the project; the occurrence of the risk causes a loss of the primary function(s) of the project. Remedial actions, even large in scope, are not enough to avoid serious damage.
V	Catastrophic: Project failure that may result in serious or even total loss of the project functions. Main project effects in the medium-long term do not materialise.

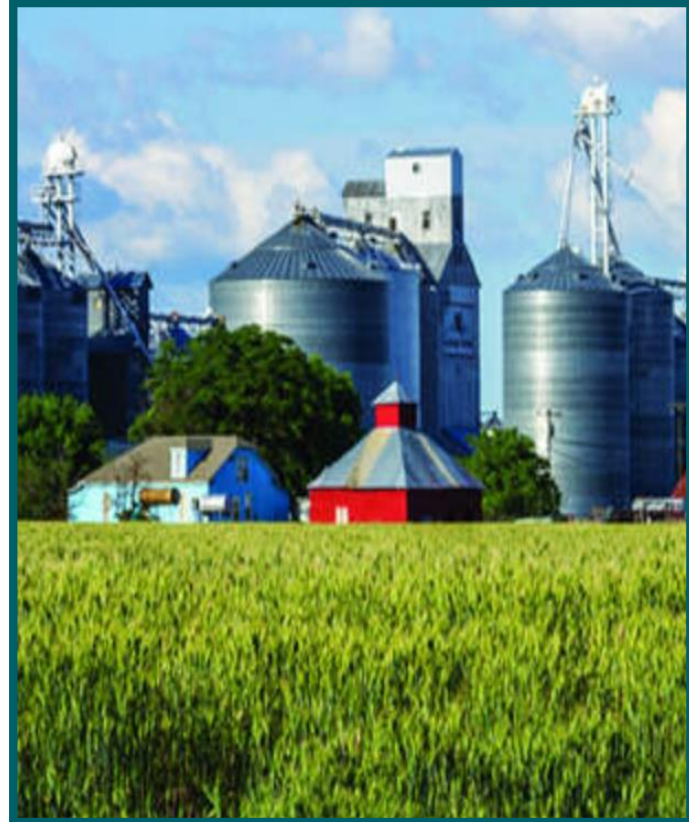
Risk matrix

Risk level	Colour	Severity / Probability	I	II	III	IV	V
Low		A	Low	Low	Low	Low	Moderate
Moderate		B	Low	Low	Moderate	Moderate	High
High		C	Low	Moderate	Moderate	High	High
Unacceptable		D	Low	Moderate	High	Very High	Very High
		E	Moderate	High	Very High	Very High	Very High

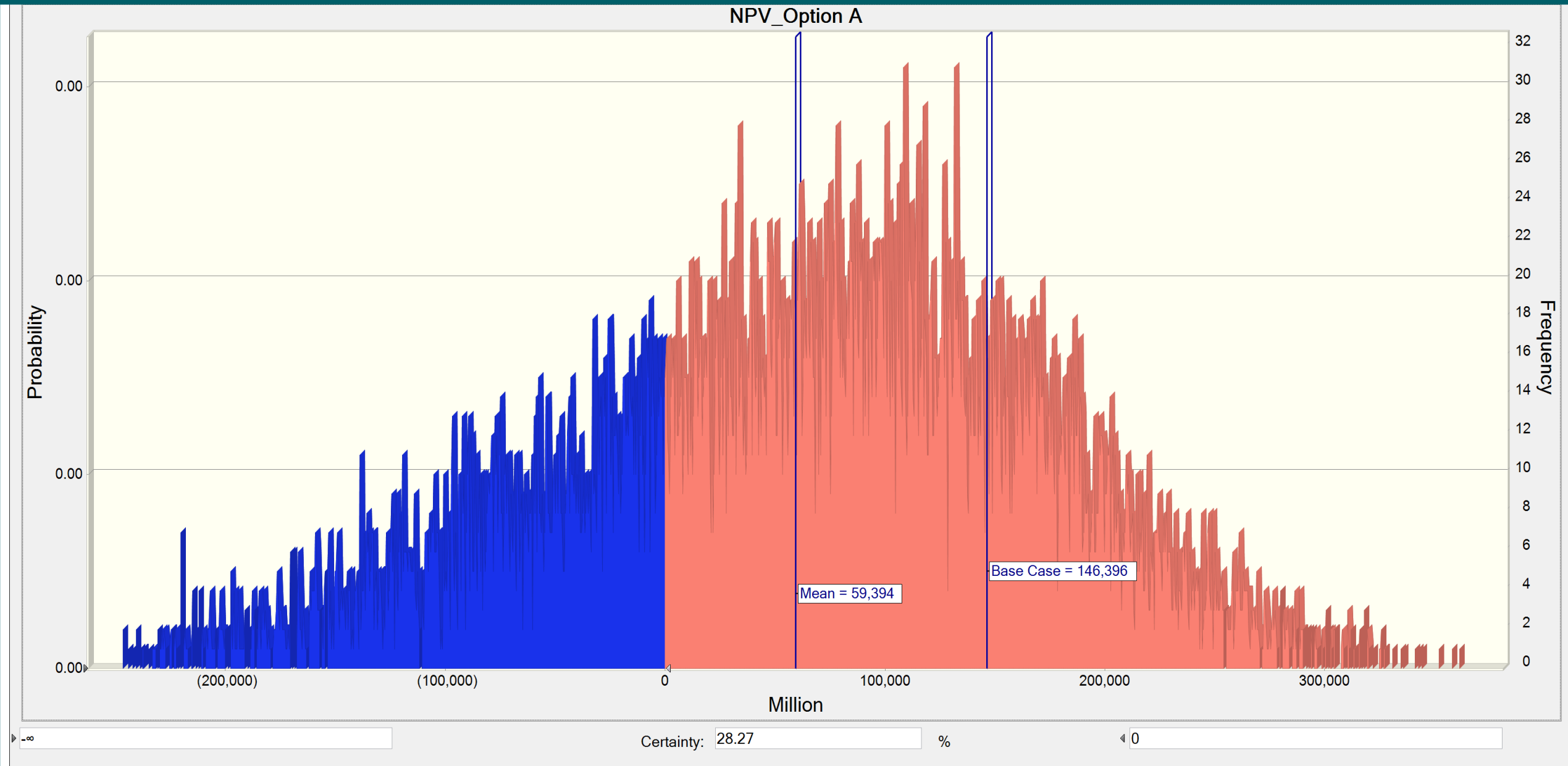
Severity / Probability	I	II	III	IV	V
A	Prevention or mitigation		Mitigation		
B					
C					
D	Prevention		Prevention and mitigation		
E					

An illustrative example

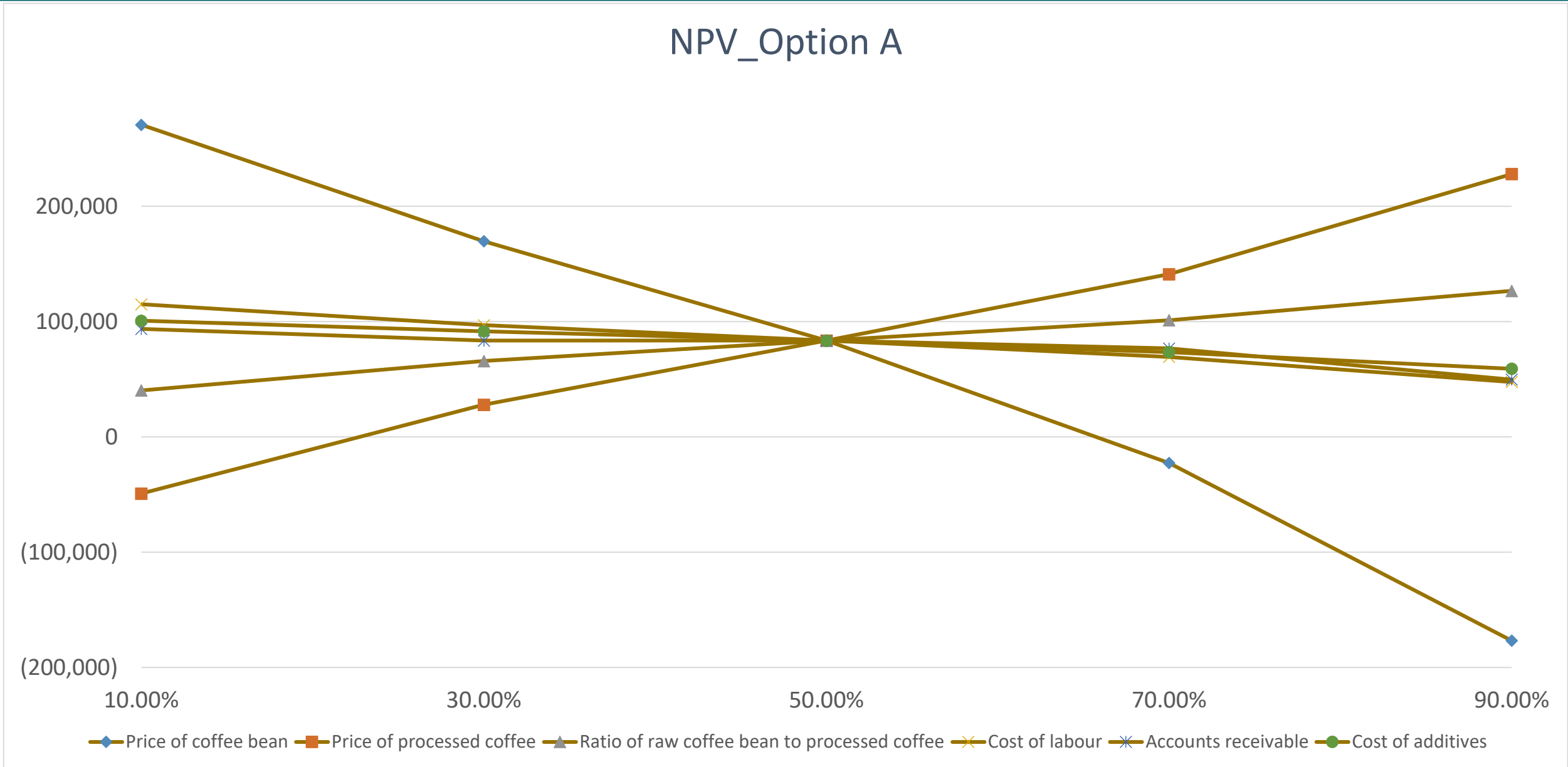
PEO project



An illustrative example

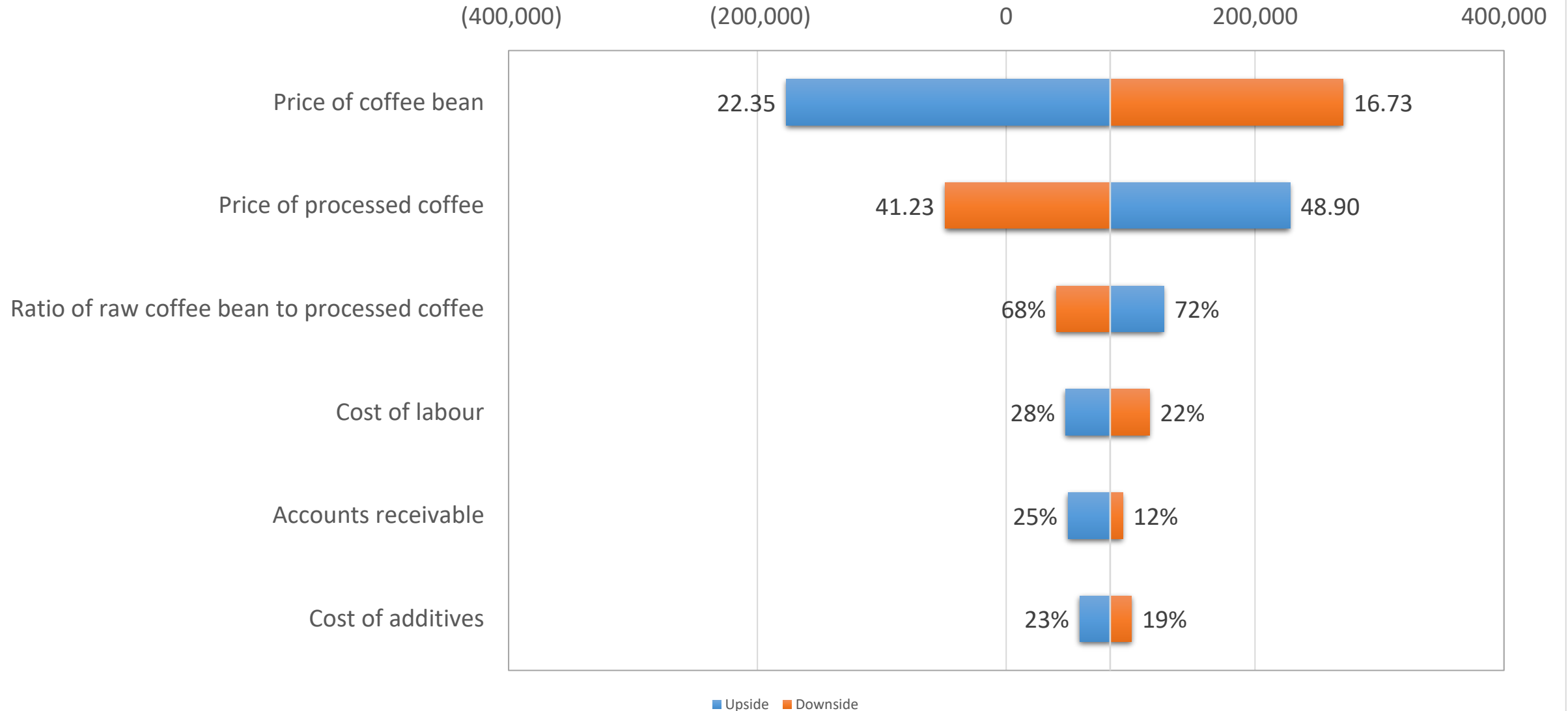


An illustrative example



An illustrative example

NPV_Option A

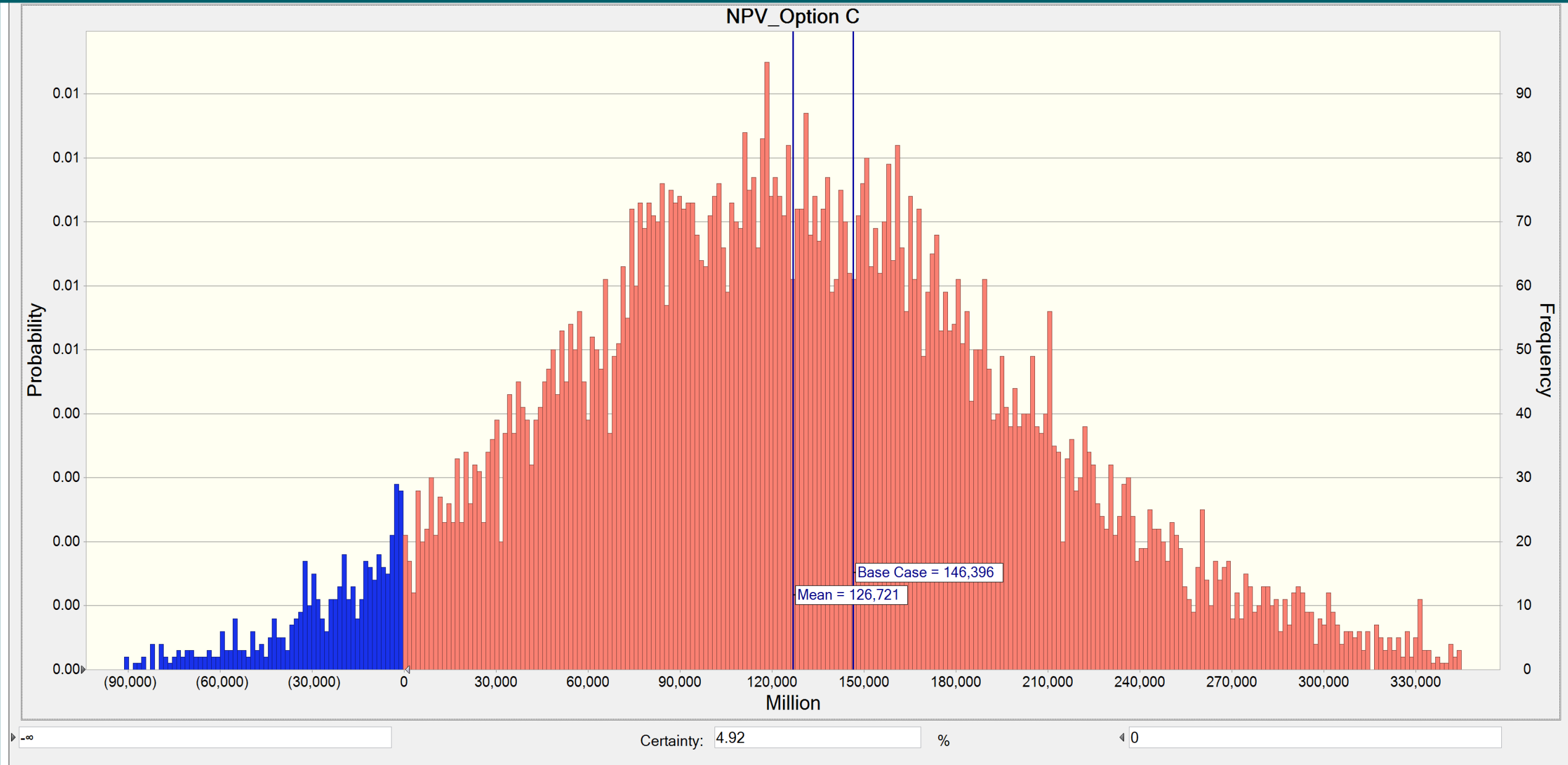


An illustrative example

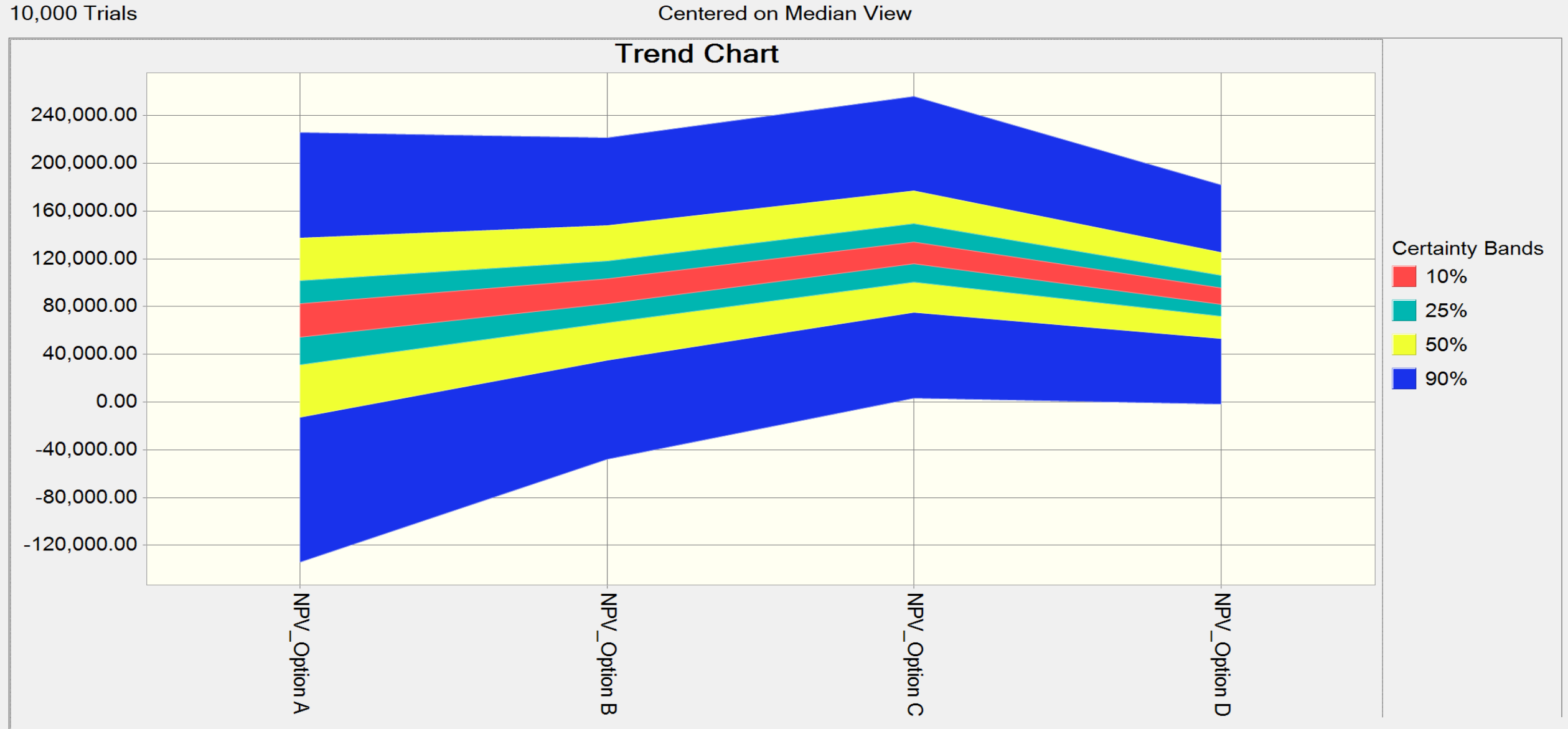
Price of coffee bean and price of processed coffee are two most important risk variables. The analyst suggests the project owners to mitigate risk by using contracts.

Option	Contract term (Coffee bean price)	Contract term (Processed coffee price)	Mean NPV	Probability [NPV < 0]
A	None	None	59.394	28.27%
B	[16 – 22]	None	89.252	13.79%
C	[17 – 20]	None	126.721	4.92%
D	[17 – 20]	[40 – 47]	86.429	5.45%

An illustrative example



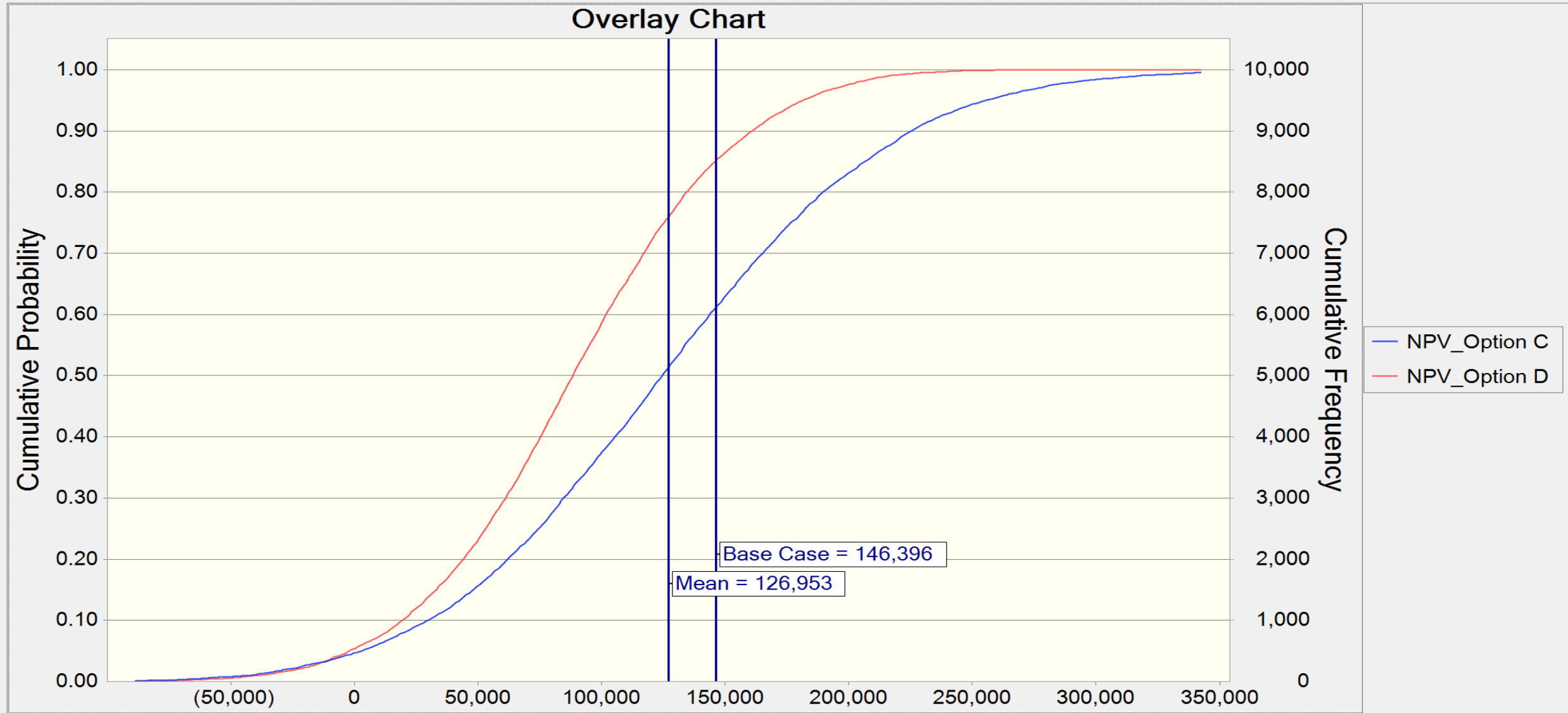
An illustrative example



An illustrative example

10,000 Trials

Cumulative Frequency View



Risk analysis

Q&A section

